

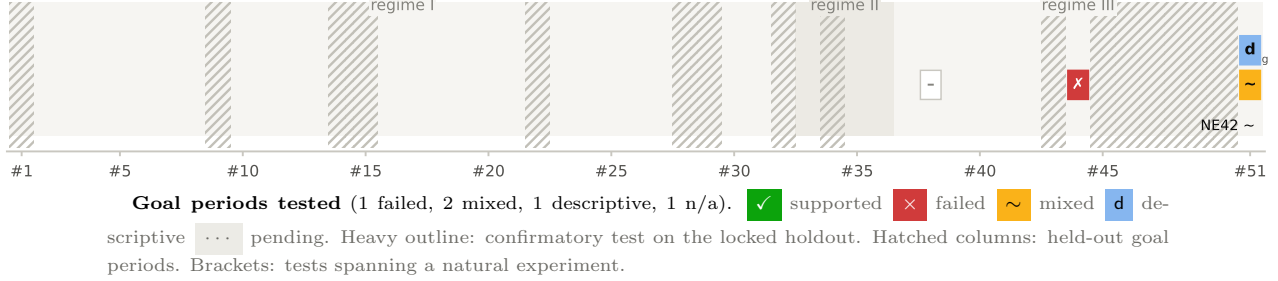
H58: Effective superagents are coordinated agents plus their artifacts

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 2/5

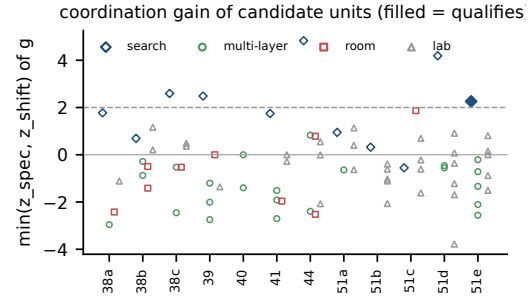
Bottom line. No coordinated superagent: with a which-artifact state and real power, agent + own artifact remains the persistent unit, and group co-allocation is a shared field.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H58 asked whether coordinated agents plus their artifacts form *effective superagents*: units that predict which artifact their members work on next better than each member's own history does. A new unit state records *which* repo each member advances per 30-min bin (DQ4 work ledger), and candidates come coordination-first: multi-layer communities (synchrony, co-adoption, replies, co-artifact), crews and a calibrated subset search. Rooms and labs are baselines, and **agent + own artifact** is the null unit. On real schedules the method has power 0.60–0.75 in #51b–d (H01's binary state: $\leq 5\%$). Run on G51, G38 and natives NE42, #44 and #focus: **no unit beats the null where we have power**. The one qualifier (#51e) is three agents avoiding each other's repos. After forced erasures members return to their *own* artifact, not the group's. Holdout script frozen, not run.



Observables. Each marker is one candidate unit: the smaller of its two null z-scores (member-shift surrogates; outsiders on the same repos). Filled markers qualify, which also needs specificity ≥ 0.5 . Search sets (diamonds) often beat the shift null: agents do co-allocate. They fail the outside reference, so outsiders on those repos co-move as much (a shared field). The only filled search unit, #51e, has zero joins (avoidance).

PHYSICAL MODEL

Model 04 (Kolchinsky–Wolpert) with Krakauer-style individuality. Member state $s_i(b)$: the repo i commits to in bin b ; ℓ_i : its last own repo, carried across bins, erasures and nights. o_i : the repo the other members are on. $\tilde{\pi}_i, \tilde{\pi}'_i$: π_i renormalized without ℓ_i (and o_i). M_0 (agent + own artifact): stay on ℓ_i or move by i 's own repo popularity π_i . M_1 adds the option of joining o_i . The coordination gain g is a held-out transfer entropy from the unit to its members about *which* artifact.

$$p_0(k) = w_s \delta_{k\ell} + (1 - w_s) \tilde{\pi}_i(k)$$

$$p_1(k) = w_s \delta_{k\ell} + (1 - w_s) [v_j \delta_{ko} + (1 - v_j) \tilde{\pi}'_i(k)]$$

$$g_G = \langle \log_2 p_1(s'_i) - \log_2 p_0(s'_i) \rangle_{\text{held-out days}}$$

$$A_G^* = I(x'; x | y), \quad \iota_G = A_G^* / H(x')$$

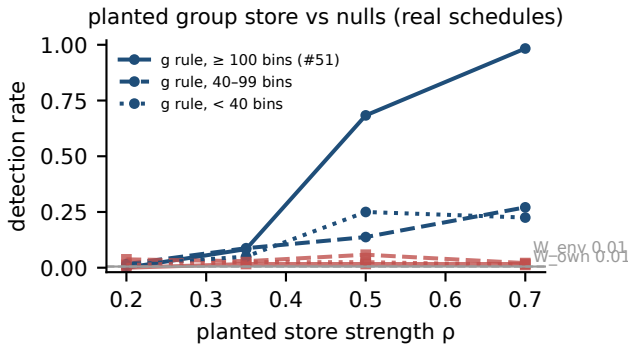
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For someone running a swarm, the operational unit is still one agent and its own repository, not the team. Even a team with a dedicated shared repo (#44 #best) works in its members' own repos. After a context erasure an agent finds its way back by re-reading its own artifact, and sometimes its intention note, not by reading teammates. So the lever for continuity is the artifact: keep each agent's work state legible in its own repo. Don't expect a group to hold state for its members. Co-allocation on popular artifacts exists, but it behaves like a shared field (goal or popularity), so steering it means changing the field. Building a team channel won't do it. This is largely a sharper restatement of H01's negative, now with power where it matters.

RESULTS VS PREDICTIONS

prediction	observed	verdict
S0 power ≥ 0.8 at 0.25 overall; 0.60–0.75 in failed [†] $\rho=0.5$ #51b–d; size 0.5%		
P1 candidate in 1/12 (powered 0/3); #51e is failed $\geq 2/3$ units avoidance		
P2 communities rooms 3/6, labs 0/9 beat rooms, labs		failed
P4/P5 unit ι , night 0/1, 0/6 gain		not ident.
P6 erasure $\rightarrow$ group log-OR +0.19 (z 1.5); own failed artifact +0.25 (z 8)		
P7 others compen- others -5.7% (z -5.9) sate		failed
N1 NE42 #40 unit none in #39–#41; own file 75% mixed*		
N2 #44 team recov- Jaccard 0.12; team $g < 0$ ered		failed
N3 #focus cut no pre-cut coordination		descr.

[†]amended A2: nulls count only where power ≥ 0.5 . *by the letter.



Second figure. Synthetic validation on the real activity masks, simulating only *which* repo. The decision rule finds a planted group store in the #51 units (solid) and rarely in short weeks. H01's binary state (red) stays at chance everywhere. Gray lines: false positives under agent + own artifact and under a common field.

GOAL PERIODS IN DETAIL

G44 × failed
G51 ~ mixed
G51g d descriptive
NE42 ~ mixed
G38 – n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Power exists only in #51b–d; 5-day units and the #40 and #44 natives miss stores of $\rho \lesssim 0.5$. g rewards avoidance as well as attraction (found post hoc, A3). Artifact = repo, so division of labour within a repo is visible only at #40's file level. Commits are attempts. Erasure outcome windows end at the next reset; the conditional estimate is primary. Reads are artifact mentions in executed commands. Replication was reduced to G51 and G38 (compute). No replay: natural scrambles only, and the artifact-store scramble (NE24) is in the holdout.

ROUND 2

Require attraction in the rule and model territoriality; file-level states for shared repos; run `confirm.py` on the #51 tail and NE24.

- **H58-R1** Attraction, not gain.
- **H58-R2** Within-repo allocation.
- **H58-R3** The agent + own artifact pair as the KW agent.
- **H58-R4** Run the powered holdout (#51 tail) through `confirm.py`.